

Emotion Across Speech and Faces

Shared Affective Mechanisms in Multimodal Foundation Models

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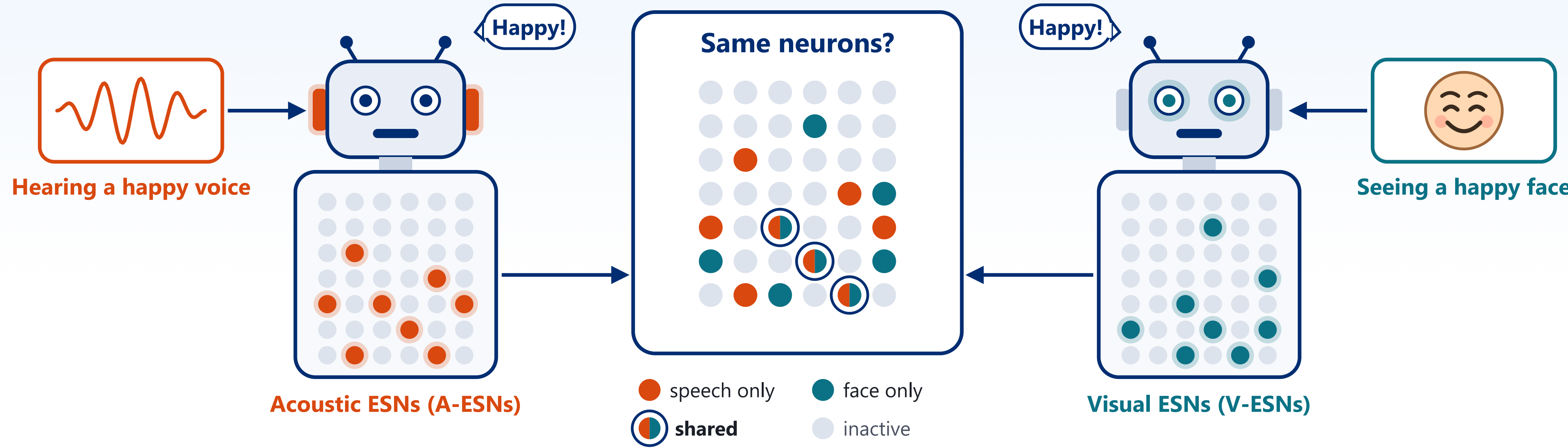
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IMPERIAL

Motivation: one emotion, two senses

Does a multimodal foundation model (MFM) recognize **emotion in a voice** and **emotion in a face** with the same internal units?



- Causal?** Does each set drive emotion recognition in its own modality?
- Overlap?** Do the two sets share neurons?
- Transfer?** Do speech neurons steer faces, and vice versa?

Method

- Collect.** Record how often each decoder MLP neuron fires on correctly recognized examples of each emotion.
- Select (ConAct).** Keep the top **0.5%** of neurons per emotion, ranked by firing margin:

$$s = p^{(1)} - p^{(2)}$$
Speech gives **A-ESNs**; faces give **V-ESNs**.
- Intervene** on the selected neurons:
deactivate: $\times 0$ | steer: $\times 1.5$
Compare against same-size random masks.

Setup

	SER	FER
Data	MSP-Podcast	AffectNet
Test / emotion	150 clips	300 images
Baseline UAR	27.5–36.9%	48.6–64.0%

Models: Gemma-4-12B-it · MiniCPM-o-4.5 · Qwen2.5-Omni-7B

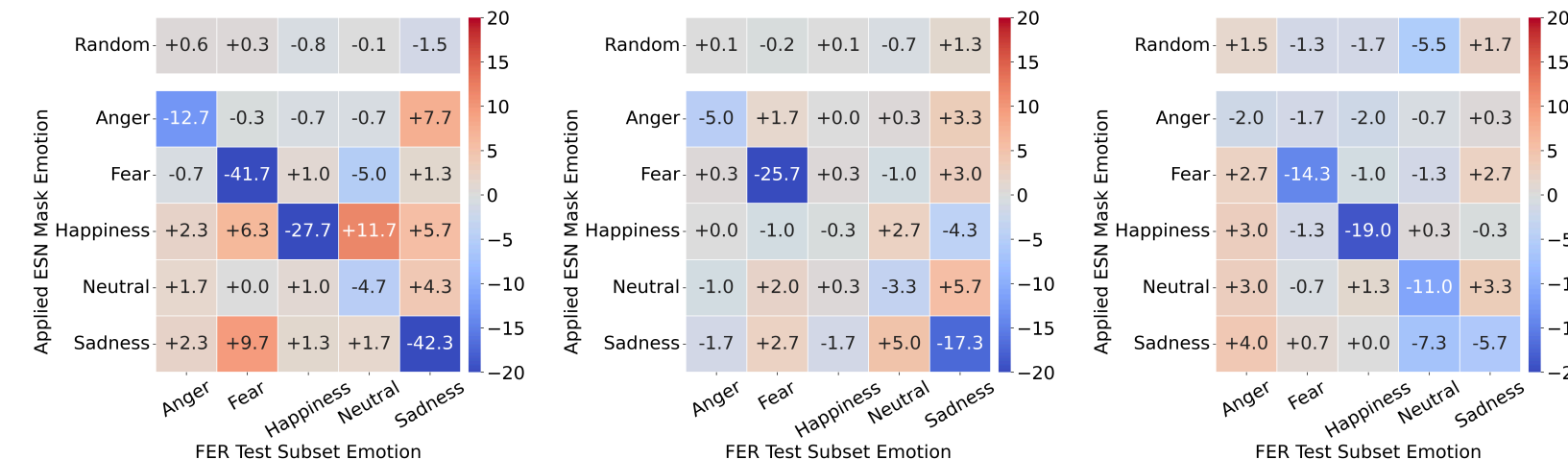
Emotions: anger · fear · happiness · neutral · sadness

Evaluation design

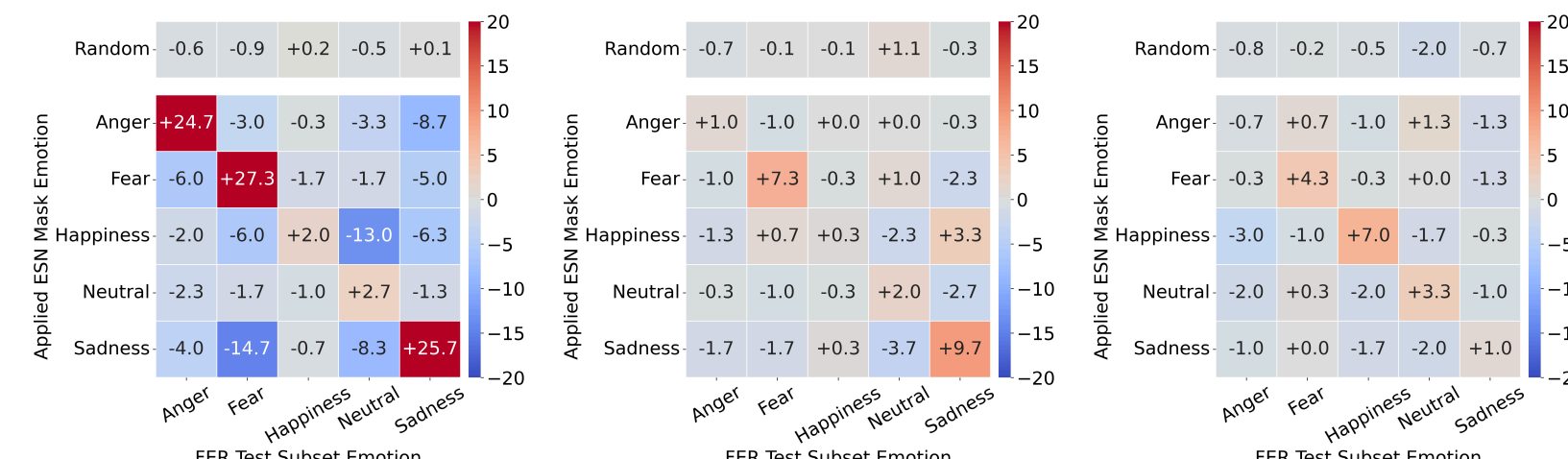
	SER (speech)	FER (faces)
A-ESN mask	within ①	cross ③
V-ESN mask	cross ③	within ①

Per-emotion FER effects

V-ESN deactivation



V-ESN steering



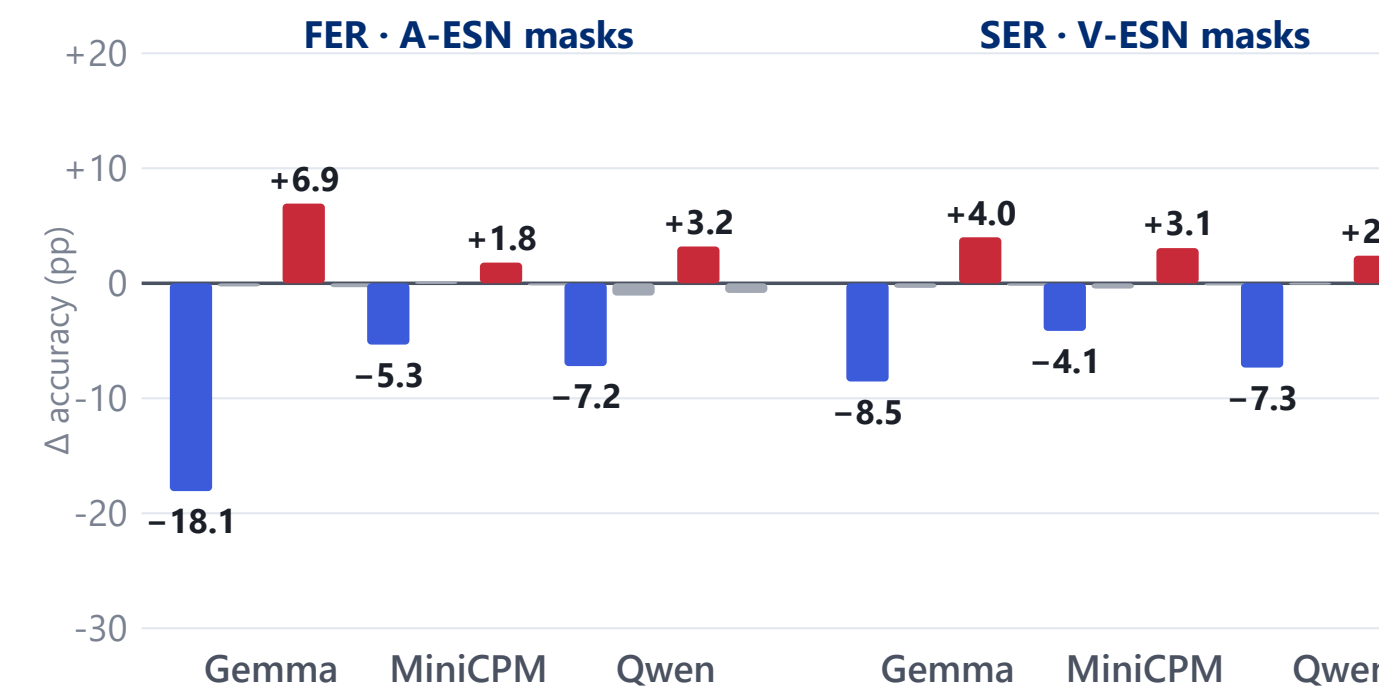
Gemma-4-12B-it MiniCPM-o-4.5 Qwen2.5-Omni-7B

Rows: mask emotion. Columns: test emotion. The effect concentrates on the **diagonal**.

③ Bidirectional transfer

Swap the masks: **A-ESNs** on faces, **V-ESNs** on speech.

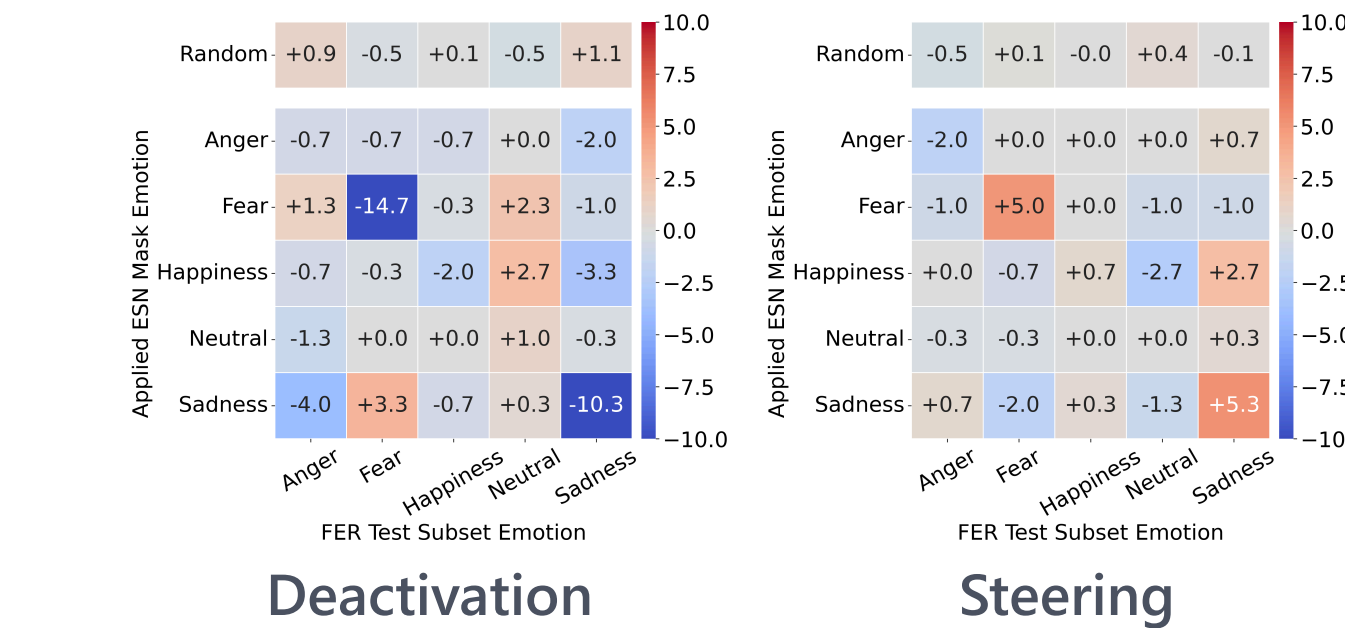
■ ESN deactivation ■ ESN steering
■ Random mask



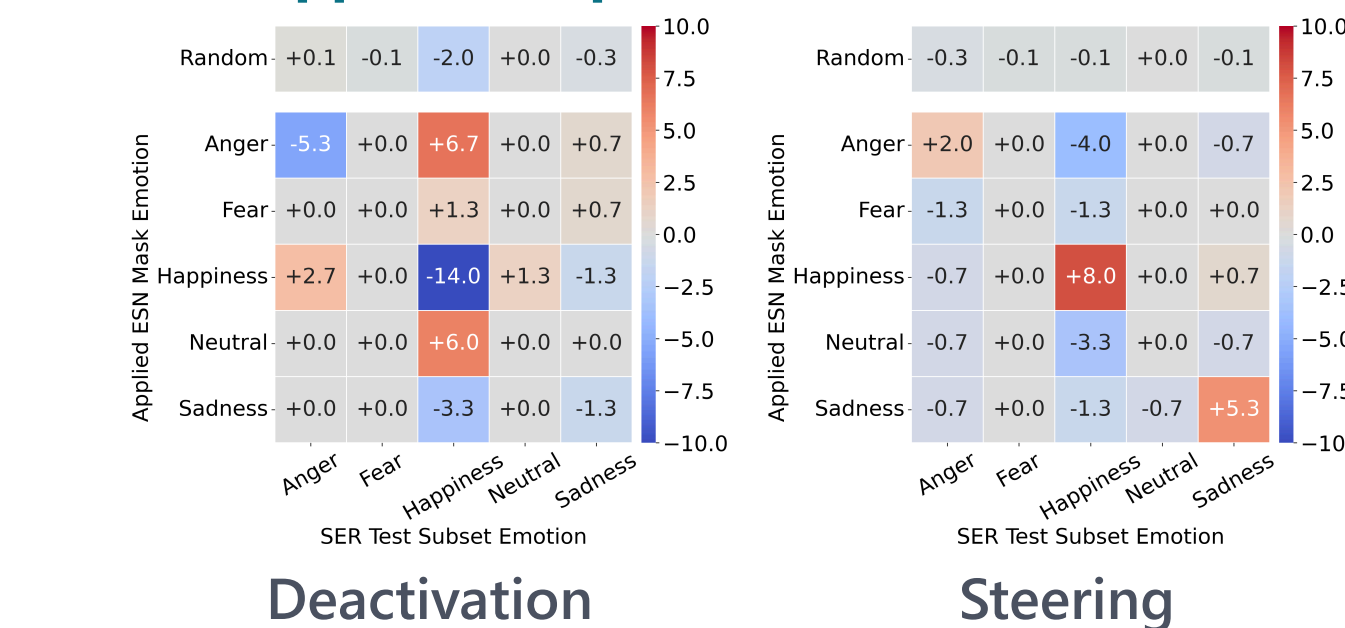
Same scale as ①. Same direction, smaller size: sharing is **partial**.

Per-emotion transfer

A-ESNs applied to faces (FER)



V-ESNs applied to speech (SER)



MiniCPM-o-4.5. The diagonal still stands out after transfer.

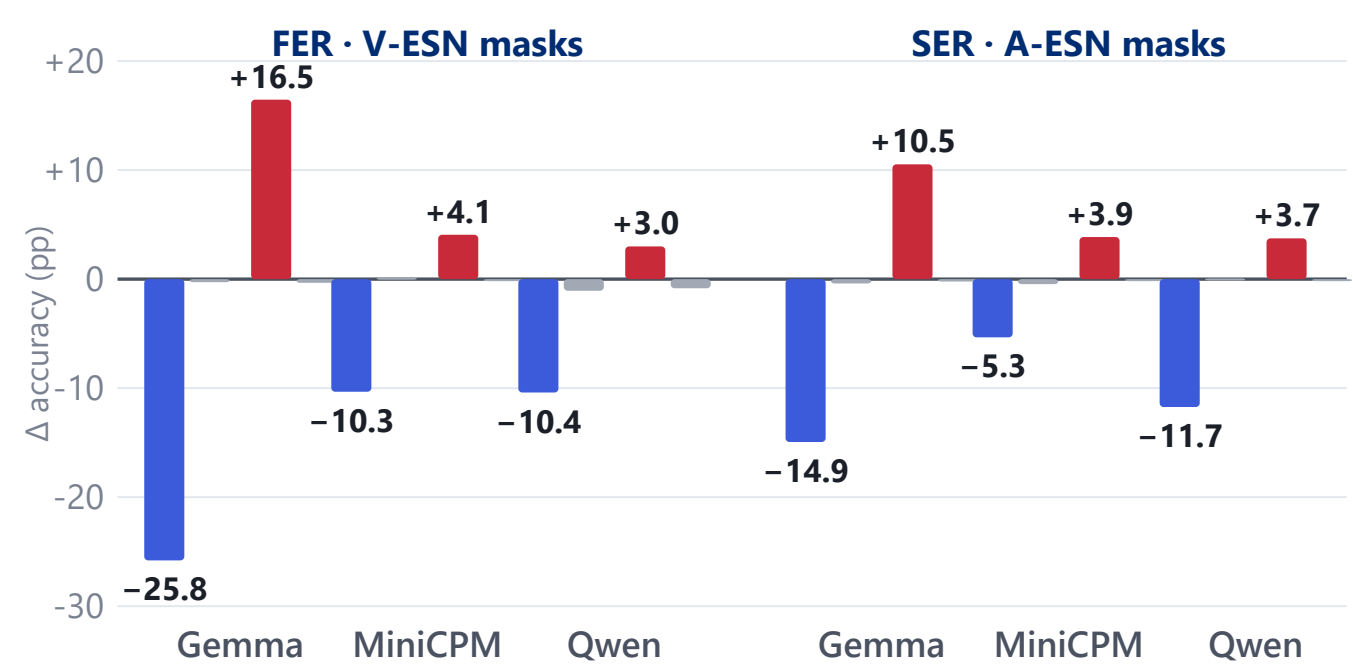
Key Findings

- Visual ESNs are causal**
Just 0.5% of neurons: masking cuts matched-emotion accuracy by up to **−25.8 pp**; steering adds up to **+16.5 pp**.
- The two sets align**
Overlap is sparse but highest for the same emotion.
- Transfer is bidirectional**
Speech neurons steer face recognition, and vice versa. Partial, but well beyond random.

① Within each modality

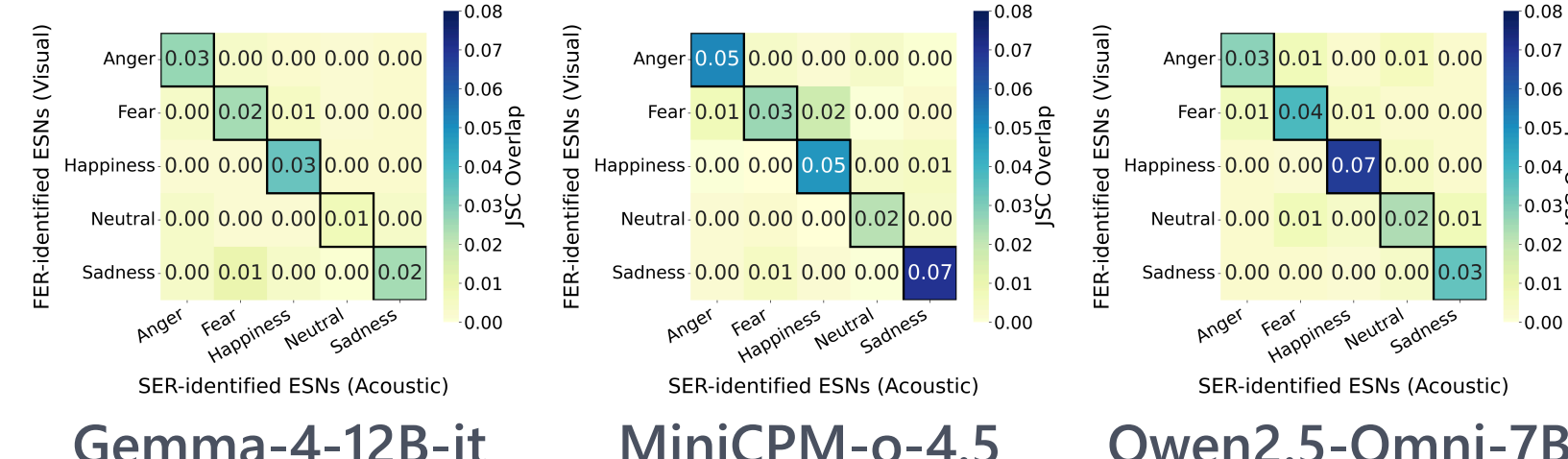
Masking an emotion's ESNs hurts *that* emotion; steering them boosts it.

■ ESN deactivation ■ ESN steering
■ Random mask (same size)



Change in matched-emotion accuracy (pp).
Random masks stay within ± 1.1 pp.

② Cross-modal alignment



Jaccard overlap: V-ESNs (rows) vs. A-ESNs (columns).

Overlap is **sparse but diagonal-dominant**: matched emotions share neurons, mismatched ones barely do.



NSF CAREER Award

ConAct: Zhao et al., "Finding Culture-Sensitive Neurons in Vision-Language Models," EACL 2026. **ESNs in LALMs:** Zhao, Schuller & Sisman, "Discovering and Causally Validating Emotion-Sensitive Neurons in Large Audio-Language Models," ACL 2026.